

BELMULLET, Ireland

WMO: 039760

Lat: 54.2275N

Lon: 10.0069W

Elev: 32

StdP: 14.68

Time Zone: 1.00 (IRL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 31.2	(c) 33.5	(d) 26.0	(e) 20.0	(f) 34.3	(g) 28.6	(h) 22.6	(i) 36.5	(j) 45.6	(k) 48.4	(l) 41.9	(m) 48.4	(n) 5.2	(o) 90	(p) 0.735

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 8.9	(c) 70.2	(d) 62.9	(e) 66.9	(f) 61.2	(g) 64.7	(h) 60.0	(i) 63.9	(j) 68.1	(k) 62.3	(l) 65.6	(m) 61.0	(n) 63.6	(o) 10.3	(p) 30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.1	83.5	65.5	60.9	79.9	63.7	59.8	76.7	62.6	29.2	68.3	28.0	65.7	27.1	63.8	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 36.1	(b) 31.6	(c) 28.4	(d) DB	(e) 27.6	(f) 75.9	(g) 2.9	(h) 3.7	(i) 25.4	(j) 78.6	(k) 23.7	(l) 80.7	(m) 22.1	(n) 82.8	(o) 19.9	(p) 85.5	(q) 74.4
			(d) WB	(e) 26.6	(f) 67.5	(g) 2.9	(h) 2.7	(i) 24.5	(j) 69.4	(k) 22.8	(l) 70.9	(m) 21.1	(n) 72.4	(o) 19.0	(p) 84.4	(q) 73.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.9	43.9	44.2	45.8	48.6	52.4	56.1	58.8	59.1	56.8	52.6	47.9	44.7
	DBStd	6.72	4.29	4.10	4.03	3.87	4.03	3.52	2.99	2.69	3.13	3.63	4.03	4.69
	HDD50	856	191	166	139	69	18	1	0	0	0	17	86	169
	HDD65	5137	654	583	597	491	391	267	196	185	246	385	514	628
	CDD50	1199	2	3	8	27	92	184	272	282	204	97	22	6
	CDD65	7	0	0	0	0	1	1	3	2	0	0	0	0
	CDH74	22	0	0	0	0	3	4	11	4	0	0	0	0
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0
(14) Wind	WSAvg	13.8	16.2	15.5	14.5	13.3	13.0	12.6	11.5	12.0	13.0	14.0	14.2	15.6
(15) Precipitation	PrecAvg	46.9	5.0	3.7	3.7	2.6	2.8	2.8	3.0	3.8	4.1	5.1	5.2	5.0
	PrecMax	57.1	9.0	9.3	6.5	5.5	5.9	5.7	5.5	9.3	8.1	10.2	9.3	10.1
	PrecMin	36.4	0.8	0.2	1.5	0.5	0.6	1.1	0.9	0.5	1.0	1.9	1.7	1.9
	PrecStd	5.5	2.0	1.9	1.3	1.0	1.3	1.0	1.1	1.9	1.7	2.1	1.6	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.6	54.3	60.2	64.7	71.9	73.7	74.8	72.7	70.0	63.8	58.0	55.5
		MCWB	51.6	51.3	53.3	56.7	61.8	64.0	65.6	65.6	63.0	59.6	55.8	53.5
	2%	DB	52.1	52.6	54.8	59.0	65.8	68.3	68.7	67.9	65.9	60.9	56.2	53.9
		MCWB	50.6	50.6	50.7	53.2	58.5	61.0	62.9	62.9	60.6	57.8	54.5	52.1
	5%	DB	51.2	51.5	52.6	56.1	61.6	64.5	65.6	65.2	63.3	59.2	54.9	52.6
		MCWB	49.7	49.7	49.7	51.6	55.7	59.0	61.2	60.9	59.4	56.6	53.2	51.0
	10%	DB	50.2	50.4	51.3	54.3	58.7	61.7	63.5	63.5	61.8	57.9	53.8	51.5
		MCWB	48.7	48.7	48.9	50.6	54.1	57.5	59.8	59.6	58.5	55.5	52.0	49.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.2	52.4	54.1	57.3	62.6	64.7	66.7	66.5	63.9	60.2	56.5	54.3
		MCDB	52.7	53.0	58.4	63.6	70.8	71.9	73.1	71.1	68.2	62.5	57.2	55.0
	2%	WB	51.2	51.3	51.9	54.2	59.0	61.8	63.7	63.7	61.9	58.6	55.0	52.6
		MCDB	51.8	52.2	53.6	57.3	64.4	67.4	67.3	66.8	64.3	60.2	55.9	53.5
	5%	WB	50.2	50.2	50.6	52.6	56.5	59.6	62.0	61.9	60.5	57.4	53.7	51.4
		MCDB	51.0	51.3	52.0	55.3	60.3	63.2	64.9	64.2	62.6	58.7	54.6	52.3
	10%	WB	49.0	49.1	49.6	51.3	54.9	58.1	60.6	60.6	59.2	56.1	52.3	50.1
		MCDB	50.1	50.3	51.1	53.8	57.8	61.0	62.9	62.5	61.3	57.7	53.6	51.3
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	8.4	9.3	10.3	10.5	9.5	8.8	8.9	9.2	8.4	7.5	7.5
		MCDBR	7.0	7.8	10.3	12.7	16.0	14.4	12.3	12.0	11.9	9.6	7.3	7.5
		MCWBR	7.4	7.4	7.9	8.7	10.1	9.3	8.1	8.2	8.4	7.8	7.4	8.0
	5% WB	MCDBR	7.1	7.0	9.3	11.0	13.7	12.6	11.1	10.1	10.2	8.5	7.0	7.2
		MCWBR	7.8	7.3	7.9	8.0	9.3	8.5	7.9	7.7	8.2	8.0	7.4	8.0
(40) Clear-Sky Solar Irradiance	taub		0.350	0.357	0.372	0.387	0.393	0.376	0.379	0.368	0.368	0.352	0.343	0.337
	taud		2.207	2.289	2.318	2.311	2.345	2.438	2.440	2.483	2.456	2.412	2.260	2.184
	Ebn at Noon		180	228	253	266	271	277	273	271	254	229	182	156
	Edh at Noon		21	26	31	36	37	34	33	30	27	23	19	17
(44) All-Sky Solar Radiation	RadAvg		183	400	765	1259	1604	1661	1477	1261	882	503	231	132
	RadStd		16	35	62	94	132	147	96	97	58	38	21	15

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page